ROLLUPOS APPLICATION USER MANUAL • AUTOMOTIVE & TELEMETRY

Blue Oval Garage

Blue Oval Garage v2.1 — 2011–2026+ Ford & Lincoln USB-C Live OBD-II Backup, FORScan Modded .abt Handoff, VIN Scanner & Deterministic As-Built Engine

Version: v2.1.1 | Build Hash: #9b769fbc | 🔒 Per-User Authorized Cartridge | Auto-Synced: 2026-10-27 02:18

1. ⚡ 60-Second Quickstart Guide

1. **Select or Add Any 2011–2026+ Ford/Lincoln (`🔵 Garage & VIN`)**
Your 2023 F-150 Lightning ER Lariat (`1FT6W1EV4PWG11434`, ⚡ BEV) is pre-seeded as Vehicle #1. Tap `+ Add Vehicle` to add any 2011–2026+ Ford or Lincoln by typing a 17-char VIN or scanning the door-jamb B-pillar sticker / windshield VIN plate with the Hybrid Camera Barcode + OCR Scanner.

2. **Capture a Live USB-C OBD-II As-Built Snapshot or Import `.abt`/`.ab` (`🛡️ Backups`)**
Plug your OBDLink EX or Vgate vLinker FS into your phone via a USB-A-to-USB-C OTG adapter, put the truck in Ready/KOER mode (`ATRV >= 12.5V`), and run a ~20s `Targeted Mod Backup` across Primary `DIAG1` (`STP33` 500k) and Secondary `DIAG2` (`STP53 + STPBR500000`) or 2011–2020 MS-CAN (`STP53 + STPBR125000`)—or import existing FORScan `.abt` (including FORScan >= 2.3.23 `G0`-encoded files) and Motorcraft `.ab` XML.

3. **Browse 183 Recipes, 21 Master Spreadsheets & Custom Lines (`🎛️ Mod Menu`)**
Browse 183 verified recipes across 6 upstream platforms (`f150`, `super_duty`, `bronco`, `explorer`, `ranger_6g`, `mustang`), download 1-click `.xlsx`/`.pdf` community master spreadsheets (Livnitup, ZooDad, DanMc85) for all 21 platforms, or use `➕ Quick-Add Custom Spreadsheet Line` to enter any forum/spreadsheet hex mod with deterministic checksum dry-run.

4. **Export Modded `.abt`, Offline `Vault .zip` & Sync to Windows FORScan (`📋 Session Plan`)**
Open the `📋 Plan` tab to generate both a line-by-line hex diff checklist and a pre-computed `G0`-encoded modded `.abt` file (guarded against stale pre-OTA `F188` firmware drift). Save or Share a deterministic `Vault .zip` archive offline (`📦 Save Vault .zip` / `📤 Share Vault .zip`), or scan the 15-minute Desktop QR PIN to stage modded `.abt` files directly into `Documents\FORScan\LightningGarage_Staging\` for FORScan's `Load All` button.

2. 🧭 Screen-by-Screen & Navigation Map

SCREEN / TAB / MODAL	WHAT IT CONTROLS & HOW TO USE IT
🔵 **Garage & VIN**	Multi-vehicle garage switcher, Hybrid Camera Barcode/OCR VIN scanner with ISO 3779 Modulo-11 auto-repair, offline + NHTSA vPIC decoder, and JSON/ZIP Vault export/import.
🎛️ **Mod Menu (183 Recipes + Custom)**	3-Layer FORScan catalog: 183 platform-filtered recipes, 21 curated community master spreadsheets (.xlsx/.pdf), and Quick-Add Custom Spreadsheet Line builder.
🧩 **Modules (87 Addresses)**	Complete Ford/Lincoln control module directory (`726`, `7D0`, `720`, `740/741`, `7E7`, `746`, etc.) with CGEA 1.2/1.3 MS-CAN (`STP53` @ 125k) vs. Gen14 `DIAG2` (`STP53` @ 500k) notes, write-risk ratings, and per-vehicle backup evidence.
🛡️ **Backups & Live USB-C**	Native USB-C OTG read-only OBD-II scanner (`ATRV >= 12.5V` gate, Quick Health / Targeted / Full Backup), offline `Vault .zip` Save/Share, time-boxed QR Phone-to-PC

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	sync, `.abt`/`.ab` importer (`G0`-codec aware), and Factory drift differ.
📋 **Session Plan & Load-All Handoff**	Generates printable line-by-line FORScan checklists, `F188`-verified pre-computed modded `.abt` files, and multi-module `.zip` handoff bundles ready for Windows `FORScan.exe` (`Load All` -> `Write`).

3. 🔬 In-Depth Feature Guide & Workflows

Native USB-C OTG Read-Only UDS Engine & Dual-Bus Switching

Uses `usb-serial-for-android` with a compiled Kotlin byte-level allowlist (`0x22`, `0x19`, `0x3E`, parked-only `0x10 01/03`, and ISO-TP `STP33`/`STP53`—physically blocking `0x2E` writes, `0x27` security unlock, and raw non-ISO-TP `STP31`). Handles multi-baud negotiation, `ATS0`/`ATS1` ISO-TP slicing, ISO 14229-2 P2* (`+5.5s`, `15s` cap) UART deadline extension on `0x78` ResponsePending frames, and `ATRV >= 12.5V` battery gating.

FORScan >= 2.3.23 `G0`-Codec Modded `.abt` Handoff, Offline `Vault .zip` & Stale Pre-OTA `F188` Guard

Exports numerically sorted pure `;Block N` `G0`-encoded `.abt` files (`01 -> G1`, `101 -> M5`) and zero-dependency IEEE 802.3 CRC-32 PKZIP `Vault .zip` bundles (`FORScan_Staging/*_MODDED.abt` + `SESSION_CHECKLIST.txt` + raw `.abt` backups) while binding every baseline to its live `F188` software strategy so stale pre-OTA files cannot overwrite newly calibrated factory bytes.

Hybrid VIN Barcode + Optical Confusable Auto-Repair & 183-Recipe Engine

Combines native `BarcodeDetector` (Code 39 / DataMatrix with automatic 18th-char `I` guard stripping), ISO 3779 Modulo-11 optical confusable repair, 183 structured recipes across 6 platforms + 21 community master spreadsheets, and deterministic Ford Modulo-256 line checksum math.

4. ⚙️ Settings, Secret Vault & Multi-User Data Isolation

SETTING KEY	FIELD DESCRIPTION	TYPE	DEFAULT
`default_vin`	Default 17-Character Vehicle VIN	`text`	`1FT6W1EV4PWG11434`
`default_trim`	Default Trim Level	`text`	`Lariat`
`default_battery`	Default Battery Pack or Engine	`text`	`Extended Range (131 kWh)`

🔒 **Storage Namespace Pattern:** `rollup::v1::u::<google-email>::app::lightning-garage::*`

📦 **Active App Storage Keys:** `__user_config__` (Managed automatically)

🔐 **Private On-Device Vault Secrets:** None required (uses Hub session)

Isolated Per-User Storage (RollupBridge) · Mobile Camera / Multimodal Capture · Android Share Sheet & CSV/JSON Export · Encrypted On-Device Secret Vault

5. 🎛️ Live UI Controls Reference & Recent Updates

Interactive Controls Detected in Current Build (`v2.1.1`):

📖 Guide | 🚗 Garage & VIN | 🎛️ Mod Menu (183) | 🧩 Modules (87) | 🛡️ Backups & Factory .ab | 📋 Session Plan (0)

🚗 Garage/VIN | 🎛️ Mod Menu | 🧩 Modules | 🛡️ Backups/.ab | 📋 Plan (0)

Pro Tips & Best Practices:

- Always put your F-150 Lightning in Ready mode (foot on brake + Start in Park so the HV DC-DC converter holds ~14.4V) before running multi-module Extended Session (`10 03`) As-Built backups.
- On STN cables like OBDLink EX, Primary HS-CAN (`DIAG1`, Pins 6 & 14 @ 500k) uses ISO-TP protocol `STP33`, Gen14 Secondary HS-CAN (`DIAG2`, Pins 3 & 11 @ 500k) uses `STP53 + STPBR500000`, and 2011–2020 CGEA 1.2/1.3 MS-CAN (Pins 3 & 11 @ 125k) uses `STP53 + STPBR125000`.

Recent Cartridge Update Log (Auto-Tracked on Publish):

- `71b73ea` (2026-10-27) — RollupOS v1.2.0 + Blue Oval Garage v2.1.0: USB-C OTG UDS read engine, STP33/STP53 ISO-TP & NRC 0x78 fixes, G0 .abt handoff, PC FORScan sync
- `153cf21` (2026-10-26) — feat(gravity): sync HTTPS tunnel & mobile UX to GitHub Pages (2026-10-26 23:32)